## 1.1 DISTANCE, DISPLACEMENT, SPEED AND VELOCITY

### LEARNING GOALS

Students will:

- Review the definitions of distance displacement position, speed and velocity.
- Perform vector and/or scalar calculations on the above quantities.

### REVIEW OF MOTION CONCEPTS

### THE FLYING BALL

Vince throws a ball straight up. It rises to its maximum height and then Simeon catches it 1.2 m above Vince's hand on its way back down.

*List all the known information about the ball's motion. If the info is not given, leave it blank*

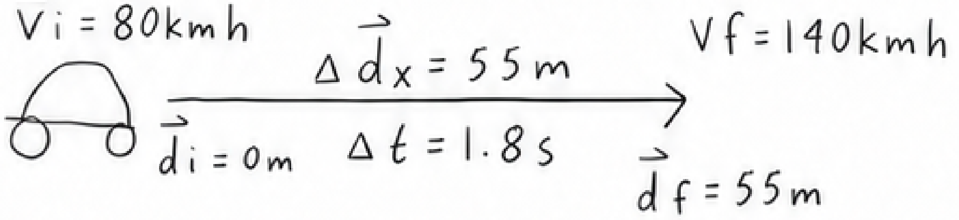
| | | | |
|---|---|---|---|
| $d$ | distance | $\vec{a}$ | acceleration $\vec{a}_x = 0 m/s^2$ $\vec{a}_y = -9.8 m/s^2$ |
| $\vec{d}$ | positian At catch: +1.2m[up] | $\vec{v}_i$ | initial velocity |
| $\Delta\vec{d}$ | displacement $\Delta\vec{d} = +1.2m$ [up] | $\vec{v}_f$ | final velocity |
| $\vec{v}_{av}$ | overage velocity | $\Delta\vec{v}$ | change in velocitg |

### THE DIFFERENCE BETWEEN $\vec{v}_{av}$ AND $\Delta\vec{v}$

A Porsche Carrera accelerates from 80.0 km/h to 140. km/h in 1.8 s and covers 55 m.

$- \longleftrightarrow +$

*Find $\vec{v}_{av}$ and $\Delta\vec{v}$.*

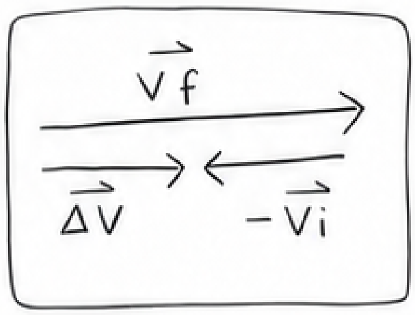


① $\vec{V}_{av} = ?$

$$\vec{V}_{av} = \frac{\Delta\vec{d}}{\Delta t}$$

$$= \frac{55m}{1.8s}$$

$$\boxed{= 31m/s} \text{ [right]}$$



② $\Delta\vec{V} = ?$

$$\Delta\vec{V} = \vec{V}_f - \vec{V}_i$$

$$\Delta\vec{V} = 140kmh - 80kmh$$

$$\Delta\vec{V} = 60kmh$$

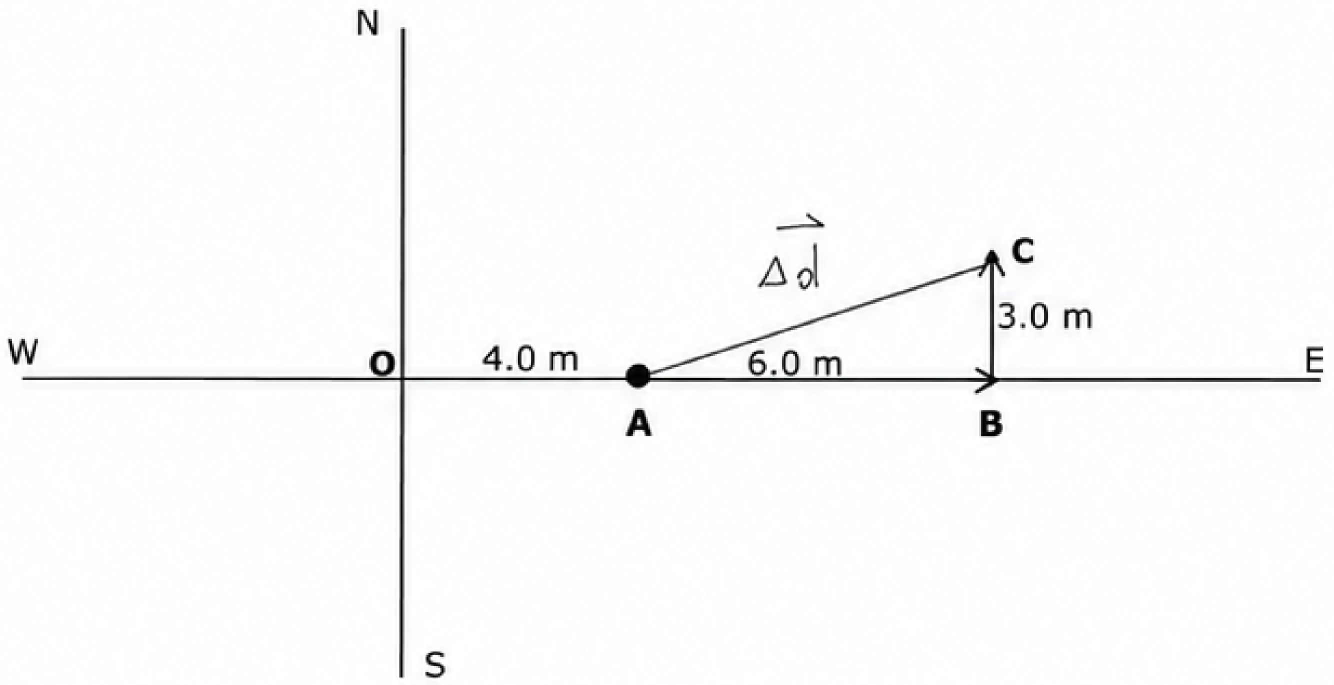
$$\Delta\vec{V} = 60kmh \times \frac{1000m}{1km} \div \frac{3600s}{1h}$$

$$\Delta\vec{V} = 60k\cancel{mh} \times \frac{1/3.6 m/s}{1k\cancel{m/h}}$$

$$\boxed{\Delta\vec{V} = 17m/s} \text{ [right]}$$

∴ Avg velocity is 31m/s [right] and the change in velocity is 17m/s [right]

## POSITION, DISPLACEMENT AND DISTANCE



In the above diagram, use the origin **O as the reference point**. A student starts at **A**, walks to **B** in 6.0 s then walks to **C** in another 6.0 s.

**POSITION**

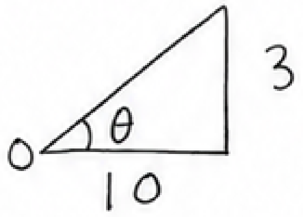
1. State the position of the student at (Answers at bottom of next page)

**A** 4.0 m [E]

**B** 10.0 m [E]

**C** (This one needs to be calculated)

10. m [E17°N]



$C' = \sqrt{10^2 + 3^2}$

$C = 10.\ m$

$\theta = \tan^{-1}\left(\frac{3}{10}\right)$ $\theta = 17°$ North of east

**DISTANCE**

2. What is the distance walked as the students walks from

**A to B** 6.0 m

**B to C** 3.0 m

**A to C** 9.0 m

**DISPLACEMENT**

3. What is the displacement as the students walks from

**A to B** $\vec{\Delta d}_{AB} = \vec{d}_B - \vec{d}_A$ . $\vec{\Delta d}_{AB} = 10\ m[E] - 4.0\ m[E]$ $\vec{\Delta d}_{AB} = 6.0\ m\ [E]$

**B to C** 3.0 m [N]

**A to C**

6.7 m [E27°N]

The displacement from **A** to **C** is the **total displacement.** You found it by adding the individual displacements from **A** to **B** and from **B** to **C**. Therefore total displacement can be found by

$$\overrightarrow{\Delta d_T} = \overrightarrow{\Delta d_1} + \overrightarrow{\Delta d_2}$$

There is another way to find total displacement. **Displacement is defined to be the change in position**. This means that displacement is given by final position minus initial position

$$\overrightarrow{\Delta d_T} = \overrightarrow{d_2} - \overrightarrow{d_1}$$

4. Try it. Take your final position at C (found above) and your initial position at **A** and do a vector subtraction. You should get the same total displacement you got above from **A** to **C**.

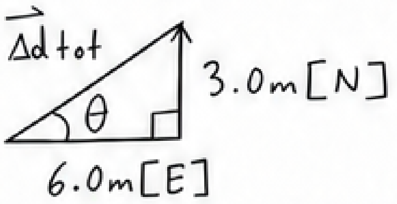
$$\Delta \vec{d}_{tot} = \vec{d}_C - \vec{d}_A$$

$$\Delta \vec{d}_{tot} = (10.0, 3.0)m - (4.0, 0)m$$

$$\Delta \vec{d}_{tot} = (10.0 - 4.0, 3.0 - 0)m$$

$$\Delta \vec{d}_{tot} = (6.0, 3.0)m$$

$$\left.\begin{array}{l} \vec{d}_C = (10.0, 3.0)m \\ \vec{d}_A = (4.0, 0)m \end{array}\right\} \text{coordinates for positions}$$

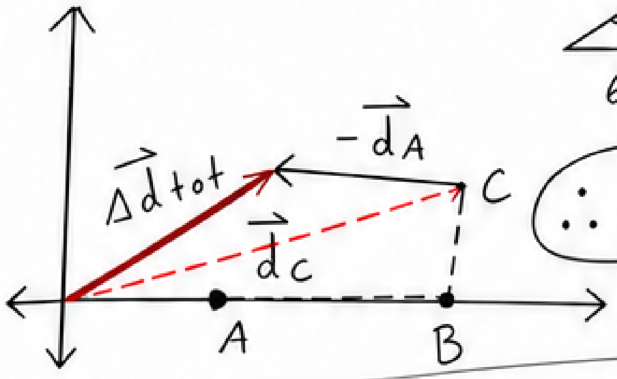


$$\Delta d_{tot} = \sqrt{3.0^2 + 6.0^2}$$

$$\Delta d_{tot} = 6.7m$$

$$\theta = \tan^{-1}\left(\frac{3}{6}\right)$$

$$\boxed{\therefore \Delta\vec{d}_{tot} = 6.7m[E27°N]} \quad \theta = 27° \text{ north of east}$$



Algebraic method point-to-point

$$\Delta\vec{d}_{tot} = \vec{d}_C - \vec{d}_A$$

$$= \vec{O}_C - \vec{O}_A$$

$$= \vec{O}_C + AO$$

$$= A_O + O_C$$

$$= \boxed{\vec{AC}}$$

$$\Delta\vec{d}_{tot} = 6.7m[E27°N] \quad \Delta\vec{d}_{AC} = 6.7[E27°N]$$

$$\boxed{\therefore \Delta\vec{d}_{tot} = \Delta\vec{d}_{AC}}$$

## AVERAGE SPEED

5. Definition: The rate at which distance is covered. Write the equation for this:

$$V_{avg} = \frac{\Delta d}{\Delta t}$$

6. Find the average speed of the student when walking from (Answers at bottom)

**A** to **B** $\frac{6.0\,m}{6\,s} = 1\,m/s$

**B** to **C** $\frac{3\,m}{63} = 0.50\,m/s$

**A** to **C** $\frac{9\,m}{12\,s} = 0.75\,m/s$

## AVERAGE VELOCITY

7. Definition: The rate of change of **position**. Write the equation for this:

$$\vec{V}_{av} = \frac{\vec{\Delta d}}{\Delta t}$$

8. Find the average velocity of the student when walking from

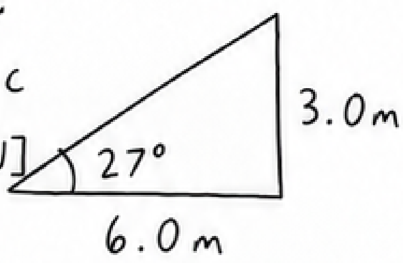
**A** to **B** $= 6m\,[E] / 6\,s \quad = 1.0\,m/s\,[E]$

**B** to **C** $= 3m\,[N] / 6s \quad = 0.50\,m/s\,[N]$

**A** to **C**

$\vec{\Delta d}_{AC}$

$= 6.7m\,[E27°N]$



$$= \frac{6.7m\,[E27°N]}{12s}$$

$$= 0.5590\,m/s\,[E\,27°N]$$

$$\vec{V}_{avAC} = 0.56\,m/s\,[E\,27°N]$$

9. What do you think it means when a definition uses the word "rate"?

rates measure the change in quantity per unit of time.

## PRACTICE

Text p 10 #1,3 – 5, p 16 #1, p 26 #1, p 28 #1 - 3

Answers for Notes

1. **A** 4.0 m[E] **B** 10.0 m[E] **C** 10.4 m[E17°N]
2. 6.0 m 3.0 m, 9.0 m
3. 6.0 m [E] 3.0 m [N] 6.7 m [E27°N]

6. 1.0 m/s 0.50 m/s, 0.75 m/s
8. 1.0 m/s [E] 0.50 m/s [N] 0.56 m/s [E27°N]